Electronic Commerce Research and Applications 62 (2023) 101317

Contents lists available at ScienceDirect

Electronic Commerce Research and Applications

journal homepage: www.elsevier.com/locate/elerap

Concrete or abstract: How chatbot response styles influence customer satisfaction

Yimin Zhu[a], Jiemin Zhang[b,*], Jiaming Liang[a]

[a] School of Business, Sun Yat-sen University, Guangzhou, China
[b] Business School, Hohai University, Nanjing, China

ARTICLE INFO

Keywords:
Chatbots
Service problem-solving
Social response theory
Response styles
Empathic accuracy
Cuteness

ABSTRACT

Lack of empathy is the key reason why chatbots cannot effectively handle service problems. Empathy consists of affective and cognitive components. Existing studies primarily emphasize empathic concern—the affective component of empathy—to improve the problem-solving performance of chatbots but neglect the cognitive component of empathy, or empathic accuracy. Focusing on empathy's cognitive component, three experiments were conducted to explore how problem response styles of chatbots (concrete vs. abstract) influence customer satisfaction based on social response theory. When chatbots respond to service problems, their concrete responses can improve customer satisfaction than abstract responses. Empathic accuracy mediates the relationship between chatbot response styles and customer satisfaction. When responses are abstract, increasing chatbots' cuteness can alleviate the negative effect of lack of empathic accuracy on customer satisfaction. These findings provide a deeper understanding of chatbots' empathy and help enterprises improve the response performance of chatbots to deal with service problems.

1. Introduction

With the development of artificial intelligence technology, chatbots are considered to possess unprecedented commercial potential (Crolic et al., 2022; Li et al., 2021; Luo et al., 2019). Given their advantages of reducing service costs and greatly improving service efficiency, chatbots are being widely used in e-commerce service encounters (Chung et al., 2020; Zhang et al., 2023). For example, the "TrayQuest" chatbot developed by McDonald's invites customers to play interactive games and receive rewards. More than 90% of customers participate in the game, creating a new direct marketing channel.

As chatbots have the advantage of being on duty at any time, more and more enterprises are using chatbots to respond to customer service problems (Jenneboer et al., 2022). Yet even as the response speed of chatbots is far faster than that of human employees (Li et al., 2021), customers are not quite satisfied with the problem-solving performance of chatbots (Zhang et al., 2023). One of the leading factors is that chatbots lack empathy (Hu et al., 2021; Lv et al., 2022c; Zhang et al., 2023; Zhu et al., 2023). Empathy is a key quality for human employees to make contact with customers (Belanche et al., 2020). Especially in the case of service problems, empathic responses can convey human employees' understanding of customer needs, which is conducive to effectively handling service problems (Lv et al., 2022c). Relative to human, however, chatbots lack empathy (Huang and Rust, 2018). As a result, when chatbots handle service problems, their responses often fail to meet the emotional needs of customers.

Considering that chatbot adoption will grow continually in e-commerce (Li and Wang, 2023) and the importance of empathy in the service problem-solving process (Lv et al., 2022c), it is crucial to improve the empathy of chatbots, which may realize the commercial value of chatbots in the service problem-solving process. A considerable literature stream has explored how to improve the empathy of chatbots from the appearance design and language styles. The appearance design, including using a female avatar (Jones et al., 2022) and cute appearances (Lv et al., 2022c), has a positive influence on the empathy of chatbots. Language styles, such as a social-oriented style (Maar et al., 2023) and a multiple-choice communication strategy (Zhou et al., 2023), also improve the empathy of chatbots.

Though the exploration of the existing literature is valuable, one shortcoming in prior studies is that they ignore empathy involves both cognitive and affective dimensions (Decety and Meyer, 2008; Zaki et al., 2008). For chatbots, affective empathy, empathic concern, is difficult to

* Corresponding author.
E-mail address: jminzhang@hhu.edu.cn (J. Zhang).

https://doi.org/10.1016/j.elerap.2023.101317
Received 23 February 2023; Received in revised form 6 August 2023; Accepted 18 September 2023
Available online 19 September 2023
1567-4223/

acquire, whereas cognitive empathy, empathic accuracy, is an acquired ability (Decety et al., 2011). Chatbots can recognize and infer customers' emotions correctly with sufficient programming skill and training (Huang and Rust, 2018). Thus, focusing on the cognitive dimension that is easier for chatbots to improve can help us better understand the empathy of chatbots.

Social response theory proposed by Moon (2000) indicates that social cues influence user responses to information technology. Many researchers apply language style as a social cue in the chatbot literature (Li and Wang, 2023). Given that more concrete responses reveal a higher perceived accuracy of problem recognition and solving (Packard and Berger, 2021; Wei et al., 2013), response styles of chatbots (concrete vs. abstract) may affect empathic accuracy of chatbots, thus further influencing the problem-solving performance of chatbots. Therefore, this study aims to focus on the cognitive component of empathy and improve empathic concern of chatbots through response styles, which in turn can help scholars and e-commerce companies make better use of chatbots to provide more effective problem-solving services.

Based on social response theory, we perform three studies to explore the influence of response styles of chatbots on customer satisfaction and the mediating role of empathic accuracy. The results of Study 1 and Study 2 demonstrate that compared with abstract responses, concrete responses of chatbots can improve customers' empathic accuracy, thus increasing customer satisfaction when chatbots respond to service problems. Study 3 examined the boundary conditions of the effects of chatbot response styles. Cuteness is used to alleviate the negative relationship between lack of empathic accuracy brought by chatbot abstract responses and customer satisfaction.

This research seeks to make several meaningful contributions to the existing literature on chatbot: (a) This study reveals the important role of empathic concern in improving the problem-solving performance of chatbots. Empathic accuracy, as the cognitive component of empathy (Zaki et al., 2008), received little attention in chatbot literature. Focusing on empathic concern helps both researchers and enterprises improve the empathy of chatbots from the cognitive dimension more effectively. (b) Based on social response theory, this study uses response styles as social cues to improve the empathy of chatbots. When dealing with service problems, the degree of concreteness in chatbot responses positively affects empathic concern of chatbots, thus influencing chatbot response performance. The exploration of chatbot response styles (concrete vs. abstract) is helpful to better understand the efficacy of different chatbot language styles. (c) This study clarifies the boundary conditions of the effect of chatbot response styles on customer satisfaction. Cute design is introduced to reduce the negative impact of inadequate empathy accuracy on customer satisfaction when abstract responses of chatbots are provided, thus softening the negative effect of abstract responses. In summary, this study not only offers important contributions to the existing literature by providing a deeper understanding of chatbots' empathy, but also sheds light on how to improve the response performance of chatbots so as to better deal with customer service problems.

2. Literature review

2.1. The empathy of chatbots

A chatbot in e-commerce platforms is defined as "a text-based software robot that can simulate interpersonal dialogue through natural language processing" (Song et al., 2022; pp. 1). One important aim of deploying chatbots is to respond to customer service problems anytime anywhere (Chung et al., 2020). For example, Taobao, the largest online shopping APP in China, uses a chatbot called "ALIME" to deal with service problems. Chatbots have two main characteristics in appearances and communication methods. From the appearance aspect, chatbots in the e-commerce settings mainly appear in the form of avatar (Jones et al., 2022). Chatbots use humanlike or non-humanlike profile photo on behalf of them. In terms of communication methods, chatbots communicate with customers in text (Li and Wang, 2023). Although chatbots can respond to service problems quickly (Jenneboer et al., 2022), in fact, chatbot responses are not as good as expected (Lv et al., 2022c). Research shows that when handling service problems, chatbots can't respond effectively mainly due to the lack of empathy (Hu et al., 2021; Zhang et al., 2023; Zhu et al., 2023).

Empathy is one of the basic steps in dealing with service problems (Krishna et al., 2011). Empathy refers to an individual's ability to perceive and understand the emotions of others and respond appropriately (Zaki et al., 2008). Chatbots chiefly possess mechanical and analytical intelligence (Huang and Rust, 2018). They are rational and logical, unequipped with the ability to interact emotionally with customers (Longoni and Cian, 2022). Effective service problem-solving, however, requires employees to demonstrate emotional competence (Zhang et al., 2023) and treat customers with empathy (Hennig-Thurau et al., 2006).

Considering the importance of empathy in the service problem-solving process, many researchers generate great interest in how to improve the empathy of chatbots. Existing chatbot literature suggests the appearance design and language styles influence customers' perceived empathy of chatbots. In the appearance design of chatbots, anthropomorphism has been used to increase the perceived experience and perceived warmth of chatbots (Choi et al., 2021; Lv et al., 2022a), which helps improve chatbots' capacity for empathy. Female chatbots are perceived to be more authentic than male chatbots, thus increasing the empathy of chatbots (Jones et al., 2022). Studies also link cute appearance to empathy and cuteness can improve customer tolerance when chatbots response to service problems (Lv et al., 2021). In the language styles of chatbots, the social-oriented style is more effective in improve the empathy of chatbots than the task-oriented style for GenZ (Maar et al., 2023) or for older customers with better internet competency (Chattaraman et al., 2019). A multiple-choice communication strategy of chatbots can enhance users' perceived empathy compared to an open-ended strategy (Zhou et al., 2023). Introducing emotion words into the chatbot communication systems improves perceived empathy of chatbot services (Yun and park, 2022). Humorous responses from AI-bots lead to more positive customer evaluations in the low-severity failure conditions.

Despite the growing number of chatbot research related to empathy, existing research treats empathy as a concept and ignores that there are affective and cognitive components of empathy. Therefore, the following article will look at affective empathy and cognitive empathy in detail.

2.2. Empathic concern

Empathy consists of empathic concern in its affective component and empathic accuracy in its cognitive component (Zaki et al., 2008). Empathic concern, the affective component of empathy, refers to the ability to generate spontaneous surrogate emotional experience of others' feelings (Decety and Meyer, 2008). Empathic accuracy, the cognitive component of empathy, indicates whether an actor can correctly infer others' emotions (Decety and Meyer, 2008). Empathic accuracy plays an important role in AI empathy. In the development history of AI, empathic accuracy is considered to "move the state of AI from 'analytical intelligence' with rule-based learning capabilities to more advanced 'intuitive intelligence' with more experience-based thinking capabilities" (Liu-Thompkins et al., 2022; pp. 1202). Empathic accuracy facilitates the conversation process and enhances social interactions (Smith, 2006). Thus, improving empathic accuracy is crucial for chatbots to provide better customer experience in the service problem-solving process.

In addition, empathic concern is generally considered to be an essential human trait (Zaki et al., 2008) and chatbots are born without empathic concern. Different from empathic concern, empathic accuracy is an acquired ability (Decety et al., 2011). Understanding the emotions of others may be independent of experiencing them (Blair, 2005). For example, patients with a brain injury may have difficulty experiencing disgust or fear, but this does not prevent them from understanding disgust or fear in others (Zaki et al., 2008). Therefore, even if an individual's ability of empathic concern is insufficient, this individual can improve their ability of empathic accuracy through acquisition training.

Chatbots are devoid of any perceived experience that humans possess (Gray and Wegner, 2012). Their insufficiency of empathic concern cannot be essentially resolved. But the improvement of empathic accuracy can be achieved through technical means and response language. AI empathy is a model-based approach (Xiao and Ding, 2014). As long as sufficient programming skills are available, emotions can be programmed just like reasoning and cognitive abilities, enabling chatbots to infer customer emotions correctly (Huang and Rust, 2018).

2.3. Response styles: Concrete vs. Abstract

The linguistic category model (LCM) indicates that languages are organized along a spectrum of concreteness-abstractness (Semin and Fiedler, 1991). Concrete language refers to describing an entity or a behavior in a manner that is more specific and perceptible to the eye or mind (Packard and Berger, 2021). The words used in concrete language often contain verbs that describe specific actions (e.g., she waved at me. Semin and Fiedler, 1991). In contrast, abstract language uses words that contain less information about a particular situation to describe entities or actions (Semin and Fiedler, 1991), and the words usually include adjectives to describe general characteristics (e.g., she was friendly to me. Jiga-Boy et al., 2013).

When responding to service problems, the response styles — that is, the degree of concrete or abstract language — affect customers' evaluation of a response (Palese et al., 2021; Wei et al., 2013). Concrete responses are personalized responses to a specific problem raised in the dialogue. In contrast, abstract responses are generic and general, whereby the content is vaguer and does not involve specific problems raised in the dialogue (Wei et al., 2013). In the special communication situation of service problem-handling, employees should respond concretely to the problems raised by customers (Min et al., 2015). On the one hand, by giving concrete responses, employees can send a strong signal to customers that they are actively listening to customers' problems or complaints (Min et al., 2015; Packard and Berger, 2021). Improved perceived listening allows customers to feel that their complaints are being taken seriously. On the other hand, we should note that there is a positive relationship between the probability of an event and the concreteness of its description. Events that are more likely to occur are usually described in detail (Todorov et al., 2007). Concrete responses match the problems reflected by customers, making customers feel that such responses are closer to the truth (Hansen and Wanke, 2010), and moreover, enterprises are more likely to learn and improve from these problems (Burton-Jones and Grange, 2013). Therefore, compared with abstract responses, concrete responses can improve customers' perceived listening and perceived truth of responses, ultimately enhancing the problem-solving effectiveness.

2.4. Cuteness in chatbot design

Cuteness is one of the most important trends in AI design (Huang et al., 2021; Zhang et al., 2022). "Cuteness" describes the extent to which a chatbot is seen as innocent and attractive in an endearing way (Lin et al., 2021; Nenkov and Scott, 2014). Though cuteness can be divided into kindchenschema and whimsical cuteness (Nenkov and Scott, 2014), chatbot literature mainly characterizes cuteness as being akin to kindchenschema — the "baby-scheme". In chatbot design, cuteness can be achieved through appearance design and interactive design (Lv et al., 2021; Lv et al., 2022b). Appearance design mainly includes chatbot graphics like large and round eyes and a bulging cheek (Lv et al., 2021). Interactive design refers to how it communicates with others, such as the tone of voice and language styles (Caudwell and Lacey, 2019). For example, using modal particles like "wuwu" in Chinese softens the statements and gives people higher perceived cuteness (Lv et al., 2022b).

The cute design of AI positively affects customers' perceptions and attitudes toward AI (Caudwell and Lacey, 2019). Cuteness usually means being lovable or pitiable (Nenkov and Scott, 2014). AI cute design can promote intimacy between domestic robots and family users (Caudwell and Lacey, 2019). Although the ability to experience is hard to acquire for AI (Zhu et al., 2023), cuteness can improve customers' cognitive experience of service robots (Huang et al., 2021) and increase customers' willingness to use AI applications for emotional tasks (Lv et al., 2022b). In addition, cute chatbots are perceived to be weak and powerless (Lv et al., 2022b). A cute AI assistant can increase people's tenderness for the AI assistant and reduce people's expectation of its function. Thus, people have a higher tolerance for service failures of a cute AI assistant (Lv et al., 2021).

3. Research hypotheses

3.1. Response styles and customer satisfaction

Social response theory suggests that "when presented with a technology possessing a set of characteristics normally associated with humans, people respond by exhibiting social behaviors and making social attributions toward the technology" (Moon, 2003; pp. 127). Social response theory has been used in human-AI interactions to explain people's social responses towards AI (Adam et al., 2021). Song et al. (2022) introduced social response theory into the scenario of chatbot recovery and examined how service recovery types influenced recovery satisfaction. Thus, social response theory can be a powerful theoretical basis in the chatbot recovery settings.

In the field of human-chatbot interaction research, the application of social response theory is often associated with social cues (Li and Wang, 2023; Moon, 2000). There are three types of cues that chatbots usually have, that is, visual cues (e.g., humanoid profile photo), identity cues (e.g., personal names), and conversational cues (e.g., mimicking human language) (Go and Sundar, 2019). Since chatbots mainly communicate with customers online in text (Li and Wang, 2023), language styles, a kind of conversational cues, influence customer satisfaction when chatbots respond to service problems. Prior literature found that when responding to service problems, concrete responses have several advantages such as improving customers' perceived listening (Packard and Berger, 2021) and perceived truth of responses (Hansen and Wanke, 2010) over abstract responses, thus enhancing the problem-solving effectiveness. Therefore, chatbot response styles affect customer satisfaction.

Whereas the usage of concrete responses has several benefits, studies have found that human employees are more likely to use non-specific and general language in the process of problem-handling (Packard and Berger, 2021). Human language habits are cultivated and perfected from childhood, which means that it is very difficult to change their habitual aspects (Semin et al., 2005). Enterprises often must put a great deal into costly training to shift the language habits of their employees, so that they can continuously and consistently use concrete responses during the process of problem-handling.

Somewhat like humans, chatbot responses are usually not specific. As computer programs with mechanical and analytical intelligence (Huang and Rust, 2018; Luo et al., 2019), chatbots can match service problems with an existing database, extract information from the database after retrieving similar problems, and reuse pre-set responses (Graef et al., 2021). Chatbot responses tend to be more abstract and are often repetitive and standardized too. Yet again unlike humans, chatbots can be re-programmed to easily adjust their language to move from abstractness to concreteness and change the language of their communication system. Compared with humans, chatbots have strong data acquisition and the ability to analyse. Through algorithms, they can collect and sort customer information in communication more quickly and generate more personalized responses (Wien and Peluso, 2021). Therefore, taking into consideration the operational ease of changing chatbot language and the positive role of concrete responses in service problem-handling (Min et al., 2015; Packard and Berger, 2021), increasing the degree of concreteness in chatbot responses can improve customer satisfaction. Accordingly, we proposed the following hypothesis:

H1: Chatbot response styles affect customer satisfaction. Specifically, concrete responses of chatbots improve customer satisfaction compared with abstract responses.

3.2. The mediating role of empathic accuracy

Social response theory proposed that language style can be utilized as a social cue when people response to chatbots (Li and Wang, 2023). The information contained in the languages reflects the level of empathic accuracy of the subject (Zaki et al., 2008). When chatbots are trained to contain more concrete information in their responses, customers will significantly have a higher perceived accuracy of problem recognition and solving (Packard and Berger, 2021; Wei et al., 2013). Concrete responses may reflect that chatbots have a better inference of customer emotions during the problem-solving process. Therefore, response styles of chatbots (concrete vs. abstract) may affect empathic accuracy of chatbots, thus further influencing the problem-solving performance of chatbots.

After programming potential customer emotions and problems, effective communication strategies are also important for chatbots in order to convey their understanding of the information (De Ruyter and Wetzels, 2000). According to LCM, chatbots can provide concrete responses to specific problems raised by customers in the communication process, so as to show their understanding and thinking of the problems (Packard and Berger, 2021). Therefore, chatbots' concrete responses deliver higher empathic accuracy than abstract responses. The improvement of empathy reduces negative evaluations of service problems (Radu et al., 2019). As the cognitive component of empathy, improving empathic accuracy can also increase customer satisfaction. Accordingly, this study tests the following hypothesis:

H2: The effect of chatbot response styles on customer satisfaction is mediated by empathic accuracy. Specifically, compared with abstract responses, concrete responses of chatbots bring higher empathic accuracy, thus improving customer satisfaction.

3.3. The moderating role of cuteness

Setting the responses of chatbots to be more concrete is effective for enterprises in order to improve customer satisfaction with chatbot responses. However, sometimes due to the complexity of the e-commerce service encounters, chatbots may not be able to provide concrete responses to specific problems in the first time. In this case, enterprises should apply effective methods to alleviate the negative effect of a lack of empathic accuracy, due to abstract responses, on customer satisfaction.

In line with social response theory, cute design is considered to be both visual cues and conversational cues (Lv et al., 2021). Accordingly, in chatbot design, cuteness can be easily achieved through appearance design and interactive design (Lv et al., 2022b). Changing chatbot appearances like using a cute image and adding terms of endearment and modal particles to play cute is an effective and cost-saving way for e-commerce enterprises to implement. Thus, from both theoretical and practical perspectives, cuteness is introduced to moderate the influence of chatbot response styles on customer satisfaction through empathic concern. Cuteness includes the tendency to be hurt easily and to be taken care of (Nenkov and Scott, 2014). Individuals who are cute are perceived to be weak and powerless (Sherman et al., 2013). That is to say, when chatbots with higher cuteness convey abstract responses, customers may have lower ability requirements for the chatbots to respond effectively and empathize accurately. In addition, in the process of chatbot responses, the lack of empathic accuracy brought by abstract responses is inevitably regarded as a failed service recovery for the customers. Cuteness can soften the negative effects of service failure, such as mitigating users' perception of the severity of software errors in human–computer interaction (Cheng et al., 2020), overall providing a "softening effect." Compared to chatbots with low cuteness, chatbots with high cuteness compel customers to show more tenderness, maintain a lower ability expectation, and have a higher tolerance for service failure (Lv et al., 2021; Zhang et al., 2022). Therefore, although abstract responses of chatbots bring lower empathic accuracy than concrete responses, in the face of cuter chatbots, the "softening effect" of cuteness on service failures weakens the negative effect of lower empathic accuracy on customer satisfaction. As illustrated in Fig. 1, we proposed the following hypothesis for this study:

H3: The effect of chatbot response styles on customer satisfaction is moderated by cuteness (H3a). Compared with low cuteness, high cuteness weakens the negative relationship between lack of empathic accuracy brought by abstract responses of chatbots and customer satisfaction (H3b).

4. Methodology

Three scenario-based experiments were conducted to examine the above hypotheses (Fig. 1). Study 1 primarily tested the influence of chatbot response styles (concrete vs. abstract) on customer satisfaction by measuring perceived concreteness of chatbot responses (H1). Study 2 replicated the main effects of chatbot response styles (H1) by manipulating response styles and further examined the mediating roles of empathic accuracy (H2). Study 3 studied the boundary conditions and revealed the impacts of cuteness (high vs low) moderating chatbot response styles on customer satisfaction through empathic accuracy (H3). All participants were recruited for a nominal payment from Credamo, a professional Chinese survey company (https://www.credamo.com/#/).[1] We recruited different participants for three studies.

4.1. Study 1

4.1.1. Design and procedures

The purpose of Study 1 was to primarily examine whether chatbot response styles affect customer satisfaction by measuring perceived concreteness of chatbot responses (H1). 70 adult participants were recruited for a nominal payment from Credamo. After excluding 7 participants who didn't pass the attention check, 63 participants were included in this study (M_{age} = 28.74, SD = 5.36). There were 23 males

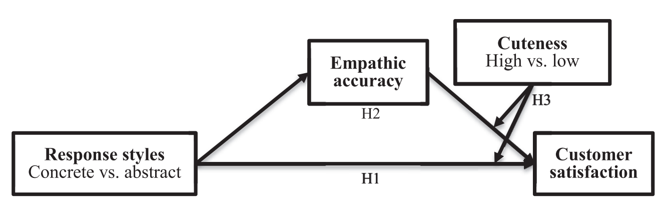

Fig. 1. Theoretical framework.

[1] The total number of domestic users in Credamo exceeds 2.8 million, covering all provincial-level regions in China and supporting hundreds of user tags. Thus, the probability of one person participating in two experiments are very small.

(36.5%) and 40 females (63.5%). 7 participants (11.1%) reported their occupations as students and 56 participants (88.9%) had a full-time job.

In Study 1, all participants were asked to imagine a hotel check-in scenario. Participants found that the room was not the one they had booked. They reflected the error of the reservation service on the hotel APP. The chatbot responded to them on the APP. Participants were shown the same response content from the chatbot. After reading the given scenario, participants completed a series of questions measuring their satisfaction with chatbot responses, and rated whether chatbot responses were concrete or abstract. Finally, we collected demographic information such as age and gender.

4.1.2. Measures

Customer satisfaction was measured by a three-item scale adapted from (Tax et al., 1998). Response styles were measured with two items adapted from Min et al. (2015) (see Table 1 for details). All items were measured in a seven-Likert scale (1 = strongly disagree; 7 = strongly agree).

4.1.3. Results

Correlation analysis results revealed a significant positive correlation relationship between perceived concreteness of chatbot responses (M = 4.75, SD = 1.76) and customer satisfaction (M = 4.78, SD = 1.57; r = 0.46, $p < 0.01$), which means the more concrete the response is, the more satisfied the participants will be.

4.1.4. Discussion

Study 1 preliminarily proved that chatbot response styles affect customer satisfaction by measuring participants' perceived concreteness of the same responses. In the next study, we manipulated chatbot response styles (concrete vs. abstract) to examine the influence of response styles on customer satisfaction again and further explored the underlying mechanism.

Table 1
Summary of measurement items.

Variables	Measurement Items (1 = Strongly disagree, 7 = Strongly agree)	α
Customer satisfaction (Tax et al., 1998)	The chatbot has done all that I expect to solve my complaint. The problem has been handled as well as it should have been. I am happy with how the chatbot handled my problem.	0.934[a], 0.959[b], 0.950[c]
Empathic accuracy (Hodges et al., 2010; Woltin et al., 2011)	The chatbot can accurately understand where the problem is coming from when responding to me. The chatbot can know exactly what I mean when responding to me. The chatbot can respond to my problem accurately.	0.953[b], 0.943[c]
Empathic concern (Gordon and Chen, 2016)	The chatbot can show its compassionate when responding to me. The chatbot can convey concerned feelings when responding to me. The chatbot feels sorry for me when I have problems.	0.849[b]
Sense of uniqueness (Longoni et al., 2019)	The chatbot did not recognize my unique circumstances. The chatbot did not tailor the response to my unique case.	0.855[b]
Response styles (Min et al., 2015)	The chatbot responses are (1 = Very generic, 7 = Very specific; 1 = Very abstract, 7 = Very concrete)	0.959[a], 0. 932[b], 0.926[c]
Perceived severity of service failure (Weun et al., 2004)	I think the service problem I met in the scenario is very severe.	–
Cuteness (Lv et al., 2021)	To what extent do you think the chatbot is cute? (1 = Not at all, 7 = Very much)	–

Notes: [a] Study 1; [b] Study 2; [c] Study 3.

4.2. Study 2

4.2.1. Design and procedures

The purpose of Study 2 was to provide a direct test of whether chatbot response styles (concrete vs. abstract) affect customer satisfaction (H1). In addition, Study 2 examined whether empathic accuracy played a mediating role in the effect of chatbot response styles on customer satisfaction (H2). A total of 140 adult participants were recruited from Credamo for a nominal payment. After excluding 11 participants who didn't pass the attention check, 129 participants were included in this study (M_{age} = 29.87, SD = 6.40). There were 64 males (49.6%) and 65 females (51.4%). 17 participants (13.2%) were students and 112 participants (86.8%) had a full-time job.

Study 2 employed a single-factor (response styles: concrete vs. abstract) between-subjects design. Participants were asked to read a hotel service problem scenario similar to Study 1. Then participants were randomly assigned to one of two conditions. Response styles were manipulated by the response content. Explanations are considered to be one of the basic elements of comprehensive apologies (Schumann, 2014).When service problems occur, providing explanations to customers can decrease the negative perceptions of the enterprise (Sparks and Fredline, 2007). Therefore, in the concrete condition, chatbots explained why the problem occurred in order to improve perceived concreteness of the response. In the abstract condition, such an explanation didn't occur and chatbots provided generic responses unrelated to the specific problem (Wei et al., 2013).

After reading the scenario, participants completed a series of questions including customer satisfaction with chatbot responses, empathic accuracy and empathic concern of chatbots, and sense of uniqueness. Sense of uniqueness refers to how different customers consider themselves from others (Longoni et al., 2019). A concern of uniqueness neglect makes customers reluctant to AI in the medical settings. Similarly, when chatbot responses are abstract, customers may be concerned that chatbots not didn't recognize the unique circumstances of their problems. Study 2 examined the potential mediating role of sense of uniqueness. Finally, demographic information was collected.

4.2.2. Measures

Customer satisfaction was measured by the same items in Study 1. As shown in Table 1, the measurement of empathic accuracy of chatbots consisted of three items adapted from Hodges et al. (2010) and Woltin et al. (2011). Empathic concern of chatbots was measured by a three-item scale revised from Gordon and Chen (2016). Sense of uniqueness was measured by two items according to Longoni et al. (2019). Perceived severity of service problem was measured by a single item ("I think the service problem I met in the scenario is very severe"; Weun et al., 2004). The manipulation check of response styles was measured in the same way as Study 1. All items were measured in a seven-Likert scale (1 = strongly disagree; 7 = strongly agree).

4.2.3. Results

The results of an independent sample t-test showed that participants in the concrete condition reported higher perceived concreteness than those in the abstract condition ($M_c = 5.82$, $M_a = 3.02$, $t = 11.12$, $p < 0.001$). The manipulation of response styles was successful.

A one-way ANOVA revealed that participants who received concrete responses including an explanation from a chatbot were more satisfied ($M_c = 5.36$, $M_a = 3.09$, $F(1, 127) = 75.67$, $p < 0.001$) than those who received abstract responses. After controlling perceived severity of service problems ($M_c = 4.89$, $M_a = 5.14$, $F(1, 127) = 1.50$, $p = 0.22$), a significant difference in customer satisfaction with different response styles was still detected ($F(1, 126) = 77.50$, $p < 0.001$), supporting H1.

Next, to examine the mediating role of empathic accuracy (H2), we conducted a bootstrapping mediation using PROCESS model 4 with 95% confidence interval and 5,000 sample sizes (Hayes, 2013). Chatbot response styles served as the independent variable; empathic concern and empathic accuracy of chatbots as mediators; customer satisfaction as the dependent variable, and perceived severity of service problems as the control variable. The results revealed that empathic accuracy of chatbots mediated the effect of response styles on customer satisfaction ($b = 1.53$, $SE = 0.27$, 95 %CI[0.99, 2.08]). Compared with abstract responses, concrete responses leaded to higher empathic accuracy of the chatbot ($M_c = 5.74$, $M_a = 3.78$, $F(1, 127) = 62.45$, $p < 0.001$), thus increasing customer satisfaction ($M_c = 5.36$, $M_a = 3.09$, $F(1, 127) = 75.67$, $p < 0.001$). In addition, the indirect effect through empathic concern of chatbots ($b = 0.04$, $SE = 0.07$, 95 %CI[-0.08, 0.19]) was not significant. Sense of uniqueness didn't show a significant indirect effect ($b = 0.10$, $SE = 0.10$, 95 %CI[-0.06, 0.32]), indicating that sense of uniqueness didn't accounted for the observed effect of response styles on customer satisfaction.

4.2.4. Discussion

Study 2 examined that when dealing with service problems, concrete responses of chatbots improved customer satisfaction than abstract responses. Though empathic concern of chatbots wasn't affected by response styles, the improvement of empathic accuracy caused by concrete responses enhanced the response performances of chatbots. In addition, this study ruled out sense of uniqueness as one possible mediator. Concrete responses show chatbots' well understanding and thinking of the problems (Packard and Berger, 2021), which may reflect that chatbots didn't ignore what customers care.

Although the positive effect of concrete responses is very significant, due to the current development level of chatbots (Huang and Rust, 2018), the response ability of chatbots may not be rapidly improved in a short time (Li et al., 2021). Therefore, in the following study, we will manipulate perceived cuteness of chatbots to reduce the negative effect of the lack of empathic accuracy due to abstract responses on customer satisfaction.

4.3. Study 3

4.3.1. Design and procedures

Study 3 aimed to examine the moderating role of cuteness (H3). A total of 250 adult participants were recruited from Credamo for a nominal payment. After excluding 16 participants who didn't pass the attention check, 234 participants were included in this study ($M_{age} = 30.59$, $SD = 7.45$). There were 108 males (46.2%) and 126 females (53.8%). The majority of participants (60.3%) had an income over 5,000 CNY per month.

Study 3 employed a 2 (response styles: concrete vs. abstract) × 2 (cuteness: high vs. low) between-subjects design. Participants were asked to imagine using the mileage to trade for goods. There was a mistake on their phone number so they didn't receive the goods. Then participants reflected the problem on the APP and a chatbot responded. Participants were randomly assigned to one of four conditions. Response styles were manipulated in the same way as Study 2. Cuteness was manipulated by changing the appearance and language of chatbots. Customers often associate roundness with cuteness, such as big round eyes and round face (Gorn et al., 2008). Terms of endearment such as "qinqin" in Chinese (similar to "honey honey") or modal particles such as "wuwu" (which indicates acting innocent) are used to play cute in the language of chatbots (Lv et al., 2021). Therefore, in the high-cuteness condition, chatbots had round eyes and faces and used words such as "qinqin" (similar to "honey honey") to play cute when communicating with customers. In the low-cuteness condition, chatbots had square eyes and faces and didn't use aforementioned words.

After reading the scenario, participants completed a series of questions including customer satisfaction with chatbot responses, empathic accuracy of chatbots and the manipulation check of cuteness. Finally, they reported their demographic information.

4.3.2. Measures

Customer satisfaction, empathic accuracy and the manipulation check of response styles were measured by the same items in Study 2. A single item was used as a manipulation check of cuteness according to Lv et al. (2021) ("To what extent do you think the chatbot is cute?", 1 = not at all, 7 = very much).

4.3.3. Results

The results of an independent sample t-test showed that participants in the concrete condition perceived the response to be more concrete than those in the abstract condition ($M_c = 4.04$, $M_a = 2.13$, $t = 9.75$, $p < 0.001$). Participants in the high-cuteness condition rated the chatbot to be cuter than those in the low-cuteness condition ($M_h = 4.41$, $M_l = 3.41$, $t = 4.34$, $p < 0.001$). The manipulation of response styles and cuteness were successful.

A 2 × 2 ANOVA revealed that customer satisfaction was significantly affected by chatbot response styles ($M_c = 4.95$, $M_a = 3.70$, $F(1, 230) = 29.34$, $p < 0.001$) and cuteness ($M_h = 4.22$, $M_l = 3.82$, $F(1, 230) = 5.45$, $p < 0.05$), nonsignificantly affected by the interaction between these two factors ($F(1, 230) = 1.87$, $p = 0.17$). H3a was not supported. Through simple effect analysis (as shown in Fig. 2), in the low-cuteness condition concrete responses leaded to higher customer satisfaction than abstract responses ($M_c = 4.39$, $M_a = 3.27$, $F(1, 115) = 23.84$, $p < 0.001$). When the cuteness of chatbots turned high, the relationship between response styles and customer satisfaction remained significant ($M_c = 4.54$, $M_a = 3.88$, $F(1, 115) = 7.86$, $p < 0.01$). However, the disparity of customer satisfaction under different levels of cuteness decreased from 1.12 to 0.66. The interaction effect between empathic accuracy and cuteness on customer satisfaction was significant ($F(1, 230) = 8.05$, $p < 0.01$).

Next, to examine the moderating role of cuteness (H3), we conducted a bootstrapping mediation using PROCESS model 15 with 95% confidence interval and 5,000 sample sizes (Hayes, 2013). Chatbot response styles served as the independent variable; empathic accuracy of chatbots as mediators; customer satisfaction as the dependent variable, and cuteness as the moderator. The index of moderated mediation, which revealed the differences between conditional indirect effects, didn't contain 0 (95 %CI[-0.83, −0.25]). In the low-cuteness condition, empathic accuracy of chatbots mediated the effect of response styles on customer satisfaction ($b = 0.62$, $SE = 0.14$, 95 %CI[0.37, 0.92]). When chatbots became cuter, the mediating effect of empathic accuracy was nonsignificant ($b = 0.10$, $SE = 0.09$, 95 %CI[−0.06, 0.31]). Thus, H3b was supported.

4.3.4. Discussion

Though concrete responses are sometimes unavailable, the findings of Study 3 indicated that cuteness design of chatbots could mitigated the effect of response styles on customer satisfaction. High-cuteness weakened the negative relationship between lack of empathic accuracy brought by abstract responses of chatbots and customer satisfaction.

5. General discussion

Although the deployment of chatbots in the service industry enables customers' problems to be answered in real time (Chung et al., 2020; Jenneboer et al., 2022), the response performance of chatbots to deal with service problems is still far from satisfactory due to chatbots' lack of empathy (Lv et al., 2022c). Previous studies focus on the affective component of empathy and enhance chatbot response performance by improving empathic concern, such as generating high empathic responses (Lv et al., 2022c) or humorous responses. However, little attention has been paid to empathic accuracy, which is the cognitive component of empathy.

Based on social response theory, this study demonstrated the effect of

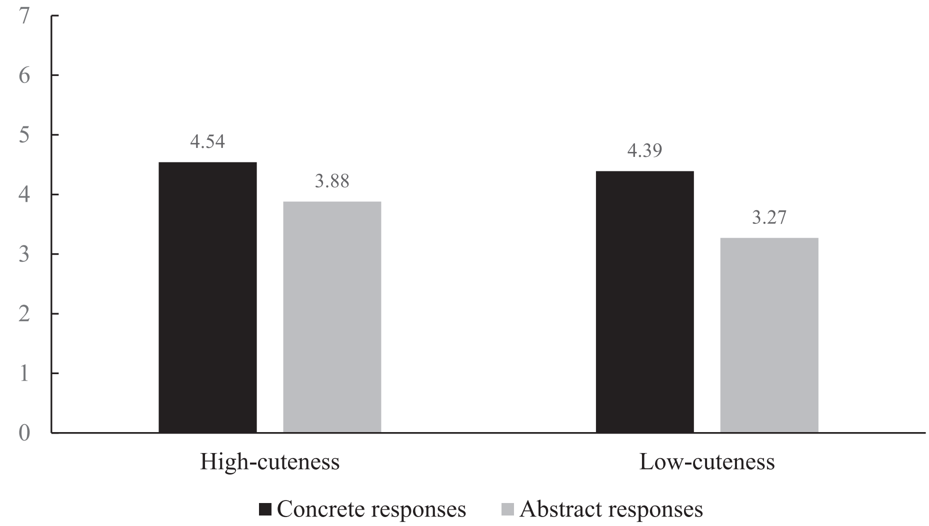

Fig. 2. The interaction effect between response styles and cuteness on customer satisfaction.

chatbot response styles (concrete vs. abstract responses) through three experiments. Concrete responses convey that employees are actively listening (Packard and Berger, 2021), and enterprises are more likely to learn from these problems and improve their operations (Burton-Jones and Grange, 2013). Concrete responses can improve customer satisfaction compared with abstract responses when chatbots respond to service problems (Study 1). Empathic accuracy plays a mediating role in the influence of chatbots' response styles on customer satisfaction (Study 2). Empathic accuracy is independent from empathic concern (Blair, 2005) and can be acquired through various means (Decety et al., 2011). Concrete responses of chatbots are perceived to have higher empathic accuracy than abstract responses, thus improving customer satisfaction. Similarly, cute chatbots are perceived to be weak and powerless (Lv et al., 2022b) and soften the negative effects of service failure (Cheng et al., 2020). Increasing the cuteness of chatbots can alleviate the negative impact on customer satisfaction with the lack of empathic accuracy caused by chatbot abstract responses (Study 3). These findings can serve as a foundation for future research to improve the response performance of chatbots in handling service problems.

5.1. Theoretical contributions

Concrete responses lead to higher customer satisfaction in the cases of both humans and chatbots (Packard and Berger, 2021). Yet compared with human employees, chatbots are readily equipped to alter their language (Semin et al., 2005). Chatbots are also more effective in information collection and retrieval (Wien and Peluso, 2021). Thus, it is more convenient for enterprises to shift the responses of chatbots to be concrete. We conclude the theoretical contribution of this study through the following aspects. First, this study reveals the psychological mechanism of how chatbot response styles influence customer satisfaction based on the cognitive component of empathy. In the field of psychology, empathy includes the affective component of empathic concern and the cognitive component of empathic accuracy (Zaki et al., 2008). Most existing literature focuses on empathic concern (Lv et al., 2022c). For example, chatbots can increase their perceived experience and perceived warmth through anthropomorphism, so as to improve their problem-handling performance (Lv et al., 2022c). But the ability to understand emotions is independent of the ability to experience emotions (Blair, 2005). This means that chatbots are more likely to have, and even improve, their empathic accuracy ability than that of empathic worry. Focusing on the mediating role of empathic concern is helpful for researchers and enterprises to understand the empathy ability of chatbots more comprehensively. Meanwhile, by exploring people's perceptions of empathic accuracy of chatbots, this study extends the application of empathic concern from human psychology research to chatbot research, enriching the scope of empathic accuracy.

Second, based on LCM, this study divides response styles of chatbots into concrete and abstract categories, exploring the impact of response styles on customer satisfaction and their psychological mechanisms. In interpersonal communication, concrete or abstract language affects communication effects (De Angelis et al., 2017; Packard and Berger, 2021). As customers gradually regard chatbots as a social presence (van Doorn et al., 2017), human–machine communication will come to more resemble interpersonal communication. When dealing with service problems, the degree of concreteness in chatbot responses has a positive impact on their response performance. In responding to service problems, the exploration of chatbot response styles (concrete vs. abstract) in human–machine communication fosters a better understanding of the efficacy of different chatbot language styles.

Finally, this study also matches the language styles of chatbots with appearance design to explore the role of cute design. Scholars universally recognize the positive impact of cuteness in chatbot design (Caudwell and Lacey, 2019; Lv et al., 2021; Lv et al., 2022b; Lv et al., 2022c; Zhang et al., 2022). Limited by the development of AI technology thus far (Huang and Rust, 2018), chatbots may not be able to provide concrete responses at first for a variety of reasons. This study found that cuteness can reduce the negative impact of inadequate empathy accuracy on customer satisfaction when abstract responses of chatbots are provided, thus softening the negative effect of abstract responses. In short, this study enriches research on cuteness in the field of chatbot service problem-handling.

5.2. Practical implications

Chatbot problem-handling performance affects customer evaluations of a chatbot's service quality (Chung et al., 2020). Although chatbots can provide real-time responses, how to improve the quality of their responses cannot be ignored today. From the perspective of social response theory and LCM, this study provides practical implications for using chatbots to respond to service problems more appropriately and effectively.

First, understanding the differences between concrete and abstract responses can help service enterprises better manage the interaction between chatbots and customers. The results of this study indicate that when responding to service problems, concrete responses of chatbots can improve customer satisfaction more successfully than abstract responses. In turn, enterprises can improve the problem-handling ability of chatbots by increasing the level of concreteness of their responses. The value of AI applications depends on the information in its databases. Accordingly, enterprises can upgrade the information databases behind chatbots to improve their dialogue management (DM) and natural language generation (NLG).

Second, enterprises can redesign the response language of chatbots to engender response styles that are more concrete, specific and targeted. For instance, responses with explanations are considered to be more specific (Wei et al., 2013): when answering a problem, chatbots can not only directly inform customers of the solutions, but also explain the reasons, so as to reduce customer dissatisfaction. Responses are also more positive when they include a paraphrase of the problem, thus making the response more personal and less generic (Min et al., 2015). When setting the response language of chatbots, enterprises can add into the responses appropriate and specific language cues related to the problem (e.g., "We are very sorry for the delay you noted." Instead of "We are very sorry about the problem."). Such chatbot language conveys both active listening and understanding (Packard and Berger, 2021). In addition, the accuracy of information presentation affects individual evaluations and judgments of the information (Kim et al., 2021). Chatbots have outstanding computational and analytical capabilities (Huang and Rust, 2018). In this way, chatbots can improve concreteness by increasing the preciseness of their response content. When a response includes numbers, for example, it is more effective for chatbots to be able to send a message in a precise informational format (e.g., "We'll get the package back to you in 3.5 h," instead of, "We'll get the package back to you in half a day").

Third, when chatbots cannot provide concrete responses, strengthening the cute attributes of chatbots can reduce customer dissatisfaction. The presentation of cuteness is not only limited to the appearance design of chatbots, such as creating chatbots with a round face and big eyes, but it can also include the dialogue styles. In Chinese and some other languages, for instance, terms of endearment or modal particles which are used to play cute in chatbot responses fosters a feeling of cuteness for customers, as "woo-woo" and "cry-cry" (Argo et al., 2010; Lv et al., 2021). In addition, the use of emojis makes the chatbots appear cuter to customers (Beattie et al., 2020). When concrete responses are unavailable, chatbots can use more emojis to convey positive nuance to customers.

5.3. Limitations and directions for future research

Despite the contributions of this study, it has some limitations that can be explored in future research. Although this study highlights the positive effect of improving the concreteness of response styles, this effect may not be consistent in the case of customer anger. In an angry state, customers typically reject chatbots, especially anthropomorphic chatbots (Crolic et al., 2022). The emotional state of any given customer, therefore, may affect the impact of response styles. Furthermore, cuteness in this study is characterized as being akin to kindchenschema. In fact, cuteness can be divided into kindchenschema and whimsical cuteness (Nenkov and Scott, 2014). Different cuteness types may affect customer behaviours. For example, compared with kindchenschema, whimsical cuteness has more of an effect on male customers (Zhang et al., 2022). Future research can consider how different types of cuteness affect people's perceptions and attitudes toward chatbots in service problem-handling. Finally, participants weren't limited to chatbots users in all three studies. Future research can recruit chatbot users or measure the chatbot usage experience of customers, thus further examining the effects of chatbot response styles.

CRediT authorship contribution statement

Yimin Zhu: Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization. **Jiemin Zhang:** Writing – original draft, Writing – review & editing, Methodology, Investigation, Conceptualization. **Jiaming Liang:** Writing – original draft, Methodology.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

This work was supported by Ministry of Education of the People's Republic of China, Humanities, Social Science Foundation under Grant [22YJA630124], the Natural Science Foundation of Guangdong Province under Grant [2021A1515011899; 2023A1515010573], and National Natural Science Foundation of China under Grant [72372023].

References

Adam, M., Wessel, M., Benlian, A., 2021. AI-based chatbots in customer service and their effects on user compliance. Electron Markets 31, 427–445. https://doi.org/10.1007/s12525-020-00414-7.

Argo, J.J., Popa, M., Smith, M.C., 2010. The sound of brands. J. Mark. 74 (4), 97–109. https://doi.org/10.1509/jmkg.74.4.97.

Beattie, A., Edwards, A.P., Edwards, C., 2020. A bot and a smile: interpersonal impressions of chatbots and humans using emoji in computer-mediated communication. Commun. Stud. 71 (3), 409–427. https://doi.org/10.1080/10510974.2020.1725082.

Belanche, D., Casaló, L.V., Flavián, C., Schepers, J., 2020. Service robot implementation: a theoretical framework and research agenda. Serv. Ind. J. 40 (3–4), 203–225. https://doi.org/10.1080/02642069.2019.1672666.

Blair, R.J.R., 2005. Responding to the emotions of others: Dissociating forms of empathy through the study of typical and psychiatric populations. Consciousn. Cognition 14 (4), 698–718. https://doi.org/10.1016/j.concog.2005.06.004.

Burton-Jones, A., Grange, C., 2013. From use to effective use: a representation theory perspective. Inf. Syst. Res. 24 (3), 632–658. https://doi.org/10.1287/isre.1120.0444.

Caudwell, C., Lacey, C., 2019. What do home robots want? The ambivalent power of cuteness in robotic relationships. Convergence 26 (4), 956–968. https://doi.org/10.1177/1354856519837792.

Chattaraman, V., Kwon, W.S., Gilbert, J.E., Ross, K., 2019. Should AI-Based, conversational digital assistants employ social-or task-oriented interaction style? A task-competency and reciprocity perspective for older adults. Comput. Hum. Behav. 90, 315–330. https://doi.org/10.1016/j.chb.2018.08.048.

Cheng, Y., Qiu, L., Pang, J., 2020. Effects of Avatar Cuteness on Users' Perceptions of System Errors in Anthropomorphic Interfaces. HCI in Business, Government and Organizations, Cham.

Choi, S., Mattila, A.S., Bolton, L.E., 2021. To Err Is Human(-oid): How Do Consumers React to Robot Service Failure and Recovery? J. Serv. Res. 24 (3), 354–371. https://doi.org/10.1177/1094670520978798.

Chung, M., Ko, E., Joung, H., Kim, S.J., 2020. Chatbot e-service and customer satisfaction regarding luxury brands. J. Bus. Res. 117, 587–595. https://doi.org/10.1016/j.jbusres.2018.10.004.

Crolic, C., Thomaz, F., Hadi, R., Stephen, A.T., 2022. Blame the Bot: Anthropomorphism and Anger in Customer–Chatbot Interactions. J. Mark. 86 (1), 132–148.

De Angelis, M., Tassiello, V., Amatulli, C., Costabile, M., 2017. How language abstractness affects service referral persuasiveness. J. Bus. Res. 72, 119–126. https://doi.org/10.1016/j.jbusres.2016.10.006.

De Ruyter, K., and Wetzels, M. G. J. J. o. S. R. (2000). The impact of perceived listening behavior in voice-to-voice service encounters. *2*(3), 276-284.

Decety, J., Meyer, M., 2008. From emotion resonance to empathic understanding: A social developmental neuroscience account. Dev. Psychopathol. 20 (4), 1053–1080. https://doi.org/10.1017/S0954579408000503.

Decety, J., Michalska, K.J., Kinzler, K.D., 2011. The Developmental Neuroscience of Moral Sensitivity. Emot. Rev. 3 (3), 305–307. https://doi.org/10.1177/1754073911402373.

Go, E., Sundar, S.S., 2019. Humanizing chatbots: The effects of visual, identity and conversational cues on humanness perceptions. Comput. Hum. Behav. 97, 304–316. https://doi.org/10.1016/j.chb.2019.01.020.

Gordon, A.M., Chen, S., 2016. Do you get where I'm coming from?: Perceived understanding buffers against the negative impact of conflict on relationship satisfaction. J. Pers. Soc. Psychol. 110 (2), 239–260. https://doi.org/10.1037/pspi0000039.

Gorn, G.J., Jiang, Y., Johar, G.V., 2008. Babyfaces, trait inferences, and company evaluations in a public relations crisis. J. Consum. Res. 35 (1), 36–49. https://doi.org/10.1086/529533.

Graef, R., Klier, M., Kluge, K., Zolitschka, J.F., 2021. Human-machine collaboration in online customer service - a long-term feedback-based approach. Electron. Mark. 31 (2), 319–341. https://doi.org/10.1007/s12525-020-00420-9.

Gray, K., Wegner, D.M., 2012. Feeling robots and human zombies: Mind perception and the uncanny valley. Cognition 125 (1), 125–130. https://doi.org/10.1016/j.cognition.2012.06.007.

Hansen, J., Wanke, M., 2010. Truth From Language and Truth From Fit: The Impact of Linguistic Concreteness and Level of Construal on Subjective Truth. Personal. Soc. Psychol. Bull. 36 (11), 1576–1588. https://doi.org/10.1177/0146167210386238.

Hayes, A.F., 2013. Introduction to Mediation, Moderation, and Conditional Process Analysis: A Regression-based Approach. Guilford Press, New York.

Hennig-Thurau, T., Groth, M., Paul, M., Gremler, D.D., 2006. Are All Smiles Created Equal? How Emotional Contagion and Emotional Labor Affect Service Relationships. J. Mark. 70 (3), 58–73. https://doi.org/10.1509/jmkg.70.3.058.

Hodges, S.D., Kiel, K.J., Kramer, A.D., Veach, D., Villanueva, B.R., 2010. Giving birth to empathy: the effects of similar experience on empathic accuracy, empathic concern, and perceived empathy. Pers. Soc. Psychol. Bull. 36 (3), 398–409. https://doi.org/10.1177/0146167209350326.

Hu, Y., Min, H., Su, N., 2021. How Sincere is an Apology? Recovery Satisfaction in A Robot Service Failure Context. J. Hosp. Tour. Res. 45 (6), 1022–1043. https://doi.org/10.1177/10963480211011533.

Huang, D., Chen, Q., Huang, J., Kong, S., Li, Z., 2021. Customer-robot interactions: Understanding customer experience with service robots. Int. J. Hosp. Manag. 99, 103078 https://doi.org/10.1016/j.ijhm.2021.103078.

Huang, M.H., Rust, R.T., 2018. Artificial Intelligence in Service. J. Serv. Res. 21 (2), 155–172. https://doi.org/10.1177/1094670517752459.

Jenneboer, L., Herrando, C., Constantinides, E., 2022. The Impact of Chatbots on Customer Loyalty: A Systematic Literature Review. J. Theor. Appl. Electron. Commer. Res. 17 (1), 212–229. https://doi.org/10.3390/jtaer17010011.

Jiga-Boy, G.M., Clark, A.E., Semin, G.R., 2013. Situating construal level: The function of abstractness and concreteness in social contexts. Soc. Cogn. 31 (2), 201–221.

Jones, E., Carol, L., Hancock, T., Kazandjian, B., Voorhees, C.M., 2022. Engaging the Avatar: The effects of authenticity signals during chat-based service recoveries. J. Bus. Res. 144, 703–716. https://doi.org/10.1016/j.jbusres.2022.01.012.

Kim, J., Giroux, M., Lee, J.C., 2021. When do you trust AI? The effect of number presentation detail on consumer trust and acceptance of AI recommendations. Psychol. Mark. 38 (7), 1140–1155. https://doi.org/10.1002/mar.21498.

Krishna, A., Dangayach, G.S., Jain, R., 2011. Service recovery: Literature review and research issues. J. Serv. Sci. Res. 3 (1), 71. https://doi.org/10.1007/s12927-011-0004-8.

Li, L., Lee, K.Y., Emokpae, E., Yang, S.B., 2021. What makes you continuously use chatbot services? Evidence from chinese online travel agencies. Electron. Mark. 31 (3), 575–599. https://doi.org/10.1007/s12525-020-00454-z.

Li, M., Wang, R., 2023. Chatbots in e-commerce: The effect of chatbot language style on customers' continuance usage intention and attitude toward brand. J. Retail. Consum. Serv. 71, 103209 https://doi.org/10.1016/j.jretconser.2022.103209.

Lin, Y.-T., Doong, H.-S., Eisingerich, A.B., 2021. Avatar Design of Virtual Salespeople: Mitigation of Recommendation Conflicts. J. Serv. Res. 24 (1), 141–159.

Liu-Thompkins, Y., Okazaki, S., Li, H., 2022. Artificial empathy in marketing interactions: Bridging the human-AI gap in affective and social customer experience. J. Acad. Mark. Sci. 50 (6), 1198–1218. https://doi.org/10.1007/s11747-022-00892-5.

Longoni, C., Bonezzi, A., Morewedge, C.K., 2019. Resistance to Medical Artificial Intelligence. J. Consum. Res. 46 (4), 629–650. https://doi.org/10.1093/jcr/ucz013.

Longoni, C., Cian, L., 2022. Artificial Intelligence in Utilitarian vs. Hedonic Contexts: The "Word-of-Machine" Effect. J. Mark. 86 (1), 91–108. https://doi.org/10.1177/0022242920957347.

Luo, X., Tong, S., Fang, Z., Qu, Z., 2019. Frontiers: Machines vs. Humans: The Impact of Artificial Intelligence Chatbot Disclosure on Customer Purchases. Mark. Sci. 38 (6), 937–947. https://doi.org/10.1287/mksc.2019.1192.

Lv, L., Huang, M., Huang, R., 2022a. Anthropomorphize service robots: the role of human nature traits. Serv. Ind. J. 43 (3-4), 213–237.

Lv, X., Liu, Y., Luo, J., Liu, Y., Li, C., 2021. Does a cute artificial intelligence assistant soften the blow? The impact of cuteness on customer tolerance of assistant service failure. Ann. Tour. Res. 87, 103114 https://doi.org/10.1016/j.annals.2020.103114.

Lv, X., Luo, J., Liang, Y., Liu, Y., Li, C., 2022b. Is cuteness irresistible? The impact of cuteness on customers' intentions to use AI applications. Tour. Manag. 90, 104472 https://doi.org/10.1016/j.tourman.2021.104472.

Lv, X., Yang, Y., Qin, D., Cao, X., Xu, H., 2022c. Artificial intelligence service recovery: The role of empathic response in hospitality customers' continuous usage intention. Comput. Hum. Behav. 126, 106993 https://doi.org/10.1016/j.chb.2021.106993.

Maar, D., Besson, E., Kefi, H., 2023. Fostering positive customer attitudes and usage intentions for scheduling services via chatbots. J. Serv. Manag. 34 (2), 208–230. https://doi.org/10.1108/JOSM-06-2021-0237.

Min, H., Lim, Y., Magnini, V.P., 2015. Factors affecting customer satisfaction in responses to negative online hotel reviews. Cornell Hosp. Q. 56 (2), 223–231. https://doi.org/10.1177/1938965514560014.

Moon, Y., 2000. Intimate exchanges: Using computers to elicit self-disclosure from consumers. J. Consum. Res. 26 (4), 323–339. https://doi.org/10.1086/209566.

Moon, Y., 2003. Don't blame the computer: When self-disclosure moderates the self-serving bias. J. Consum. Psychol. 13 (1–2), 125–137. https://doi.org/10.1207/S15327663JCP13-1&2_11.

Nenkov, G.Y., Scott, M.L., 2014. "So Cute I Could Eat It Up": priming effects of cute products on indulgent consumption. J. Consum. Res. 41 (2), 326–341. https://doi.org/10.1086/676581.

Packard, G., Berger, J., Inman, J.J., Stephen, A.T., 2021. How concrete language shapes customer satisfaction. J. Consum. Res. 47 (5), 787–806.

Palese, B., Piccoli, G., Lui, T.W., 2021. Effective use of online review systems: Congruent managerial responses and firm competitive performance. Int. J. Hosp. Manag. 96, 102976 https://doi.org/10.1016/j.ijhm.2021.102976.

Radu, A.G., Arli, D., Surachartkumtonkun, J., Weaven, S., Wright, O., 2019. Empathy and apology: the effectiveness of recovery strategies. Mark. Intell. Plan. 37 (4), 358–371. https://doi.org/10.1108/MIP-03-2018-0080.

Schumann, K., 2014. An affirmed self and a better apology: The effect of self-affirmation on transgressors' responses to victims. J. Exp. Soc. Psychol. 54, 89–96. https://doi.org/10.1016/j.jesp.2014.04.013.

Semin, G.R., Fiedler, K., 1991. The linguistic category model, its bases, applications and range. Eur. Rev. Soc. Psychol. 2 (1), 1–30. https://doi.org/10.1080/14792779143000006.

Semin, G.R., Higgins, T., de Montes, L.G., Estourget, Y., Valencia, J.F., 2005. Linguistic signatures of regulatory focus: How abstraction fits promotion more than prevention. J. Pers. Soc. Psychol. 89 (1), 36–45. https://doi.org/10.1037/0022-3514.89.1.36.

Sherman, G.D., Haidt, J., Iyer, R., Coan, J.A., 2013. Individual Differences in the Physical Embodiment of Care: Prosocially Oriented Women Respond to Cuteness by Becoming More Physically Careful. Emotion 13 (1), 151–158. https://doi.org/10.1037/a0029259.

Smith, A., 2006. Cognitive empathy and emotional empathy in human behavior and evolution. Psychol. Rec. 56 (1), 3–21. https://doi.org/10.1007/BF03395534.

Song, M., Du, J., Xing, X., Mou, J., 2022. Should the chatbot "save itself" or "be helped by others"? The influence of service recovery types on consumer perceptions of recovery satisfaction. Electron. Commer. Res. Appl. 55, 101199 https://doi.org/10.1016/j.elerap.2022.101199.

Sparks, B., Fredline, L., 2007. Providing an Explanation for Service Failure: Context, Content, and Customer Responses. J. Hosp. Tour. Res. 31 (2), 241–260. https://doi.org/10.1177/1096348006297292.

Tax, S.S., Brown, S.W., Chandrashekaran, M., 1998. Customer Evaluations of Service Complaint Experiences: Implications for Relationship Marketing. J. Mark. 62 (2), 60–76. https://doi.org/10.1177/002224299806200205.

Todorov, A., Goren, A., Trope, Y., 2007. Probability as a psychological distance: Construal and preferences. J. Exp. Soc. Psychol. 43 (3), 473–482. https://doi.org/10.1016/j.jesp.2006.04.002.

van Doorn, J., Mende, M., Noble, S.M., Hulland, J., Ostrom, A.L., Grewal, D., Petersen, J. A., 2017. Domo Arigato Mr. Roboto: Emergence of Automated Social Presence in Organizational Frontlines and Customers' Service Experiences. J. Serv. Res. 20 (1), 43–58. https://doi.org/10.1177/1094670516679272.

Wei, W., Miao, L., Huang, Z., 2013. Customer engagement behaviors and hotel responses. Int. J. Hosp. Manag. 33, 316–330. https://doi.org/10.1016/j.ijhm.2012.10.002.

Weun, S., Beatty, S.E., Jones, M.A., 2004. The impact of service failure severity on service recovery evaluations and post-recovery relationships. J. Service Market. 18 (2), 133–146. https://doi.org/10.1108/08876040410528737.

Wien, A.H., Peluso, A.M., 2021. Influence of human versus AI recommenders: The roles of product type and cognitive processes. J. Bus. Res. 137, 13–27. https://doi.org/10.1016/j.jbusres.2021.08.016.

Woltin, K.-A., Corneille, O., Yzerbyt, V.Y., Förster, J., 2011. Narrowing down to open up for other people's concerns: Empathic concern can be enhanced by inducing detailed processing. J. Exp. Soc. Psychol. 47 (2), 418–424. https://doi.org/10.1016/j.jesp.2010.11.006.

Xiao, L., Ding, M., 2014. Just the faces: exploring the effects of facial features in print advertising. Mark. Sci. 33 (3), 338–352. https://doi.org/10.1287/mksc.2013.0837.

Yun, J., Park, J., 2022. The effects of chatbot service recovery with emotion words on customer satisfaction, repurchase intention, and positive word-of-mouth. Front. Psychol. 13, 922503 https://doi.org/10.3389/fpsyg.2022.922503.

Zaki, J., Bolger, N., Ochsner, K., 2008. It takes two: the interpersonal nature of empathic accuracy. Psychol. Sci. 19 (4), 399–404. https://doi.org/10.1111/j.1467-9280.2008.02099.x.

Zhang, T., Feng, C., Chen, H., Xian, J., 2022. Calming the customers by AI: Investigating the role of chatbot acting-cute strategies in soothing negative customer emotions. Electron. Mark. 32 (4), 2277–2292.

Zhang, J., Zhu, Y., Wu, J., Yu-Buck, G.F., 2023. A natural apology is sincere: Understanding chatbots' performance in symbolic recovery. Int. J. Hosp. Manag. 108, 103387.

Zhou, Q., Li, B., Han, L., Jou, M., 2023. Talking to a bot or a wall? How chatbots vs. human agents affect anticipated communication quality. Comput. Hum. Behav. 143, 107674 https://doi.org/10.1016/j.chb.2023.107674.

Zhu, Y., Zhang, J., Wu, J., 2023. Who did what and when? The effect of chatbots' service recovery on customer satisfaction and revisit intention. J. Hosp. Tour. Technol. 14 (3), 416–429. https://doi.org/10.1108/jhtt-06-2021-0164.